Low-end

This is the entry-level market where users are heavily price-conscious, and are happy with performance that's a little better than that of their old graphics card.

All the cards in this segment are based on the nVidia GeForce FX 5200 chipset that nVidia has specifically targeted at the budget-gamer segment. We received around nine cards, eight of these were plain vanilla 5200 cards, with the only variation being the video memory, which was either 64 MB or 128 MB on-board. The ninth was an nVidia 5200 Ultra based card from Gainward. We took out some time to understand what makes this card low-priced, what features it has and what it misses out on.

The card is based on the nVidia NV34 chipset. Compared to its richer cousins, this is the only card which still uses the 0.15 micron manufacturing process. The nVidia reference design features a heat sink, and most manufacturers are sticking to that. A few manufacturers have provided an active cooling system, but only on the 128 MB video cards.

Since this card is targeted at the budget-gamer, the package contents are minimal. Only the Gainward, S-Media, and PixelView video cards featured dual-monitor display support.

The ASUS V9520 Magic card was a very good performer, and with only 64 MB video RAM, its scores were really good. Compared with the Jetway and the PixelView 128 MB video cards, this card did really good. The ASUS V9520 128 MB card delivered lower performance than its 64 MB counterpart—this is similar to the behaviour we see in the XFX brand. The Gainward 5200 card gives some good scores, and ties with the S-Media card in terms of scores in the pure GeForce FX 5200 category. Both these cards are second only to the Ultra version of the 5200 chip. Comfortable gaming frame rates are provided for DX 8 games, as in *Unreal Tournament*. High resolution gaming is still a bit slow since the images start framing with all the frills turned on. Image quality is quite good—this made the card earn a good score on that test.

The XFX GeForce FX 5200 is the the DirectX 9 graphics card for the budget-conscious buyer

The Jetway card is barely able to play the game at high resolutions with all the eye-candy turned on. This can be attributed to its bare minimum specifications; however, this does not bode well for this card. It seems that this card is just meant for those who want a GeForce FX label, without the real power that goes behind it. It did give comfortable frame rates at all resolutions in *Quake 3*, but was only able to run the *GunMetal* benchmark in playable frame-rates at a very low resolution, with all eye-candy, except for anti-aliasing turned off. This puts a serious question mark on whether the card is a viable solution for games coming out in the near future.

The S-Media card turned out to be a very good performer in the 5200 category giving better frame rates compared to its counterparts in all categories, and is second only to the Gainward GeForce 5200 Ultra. The Pine XFX 64MB FX 5200 is also an excellent performer. Considering that it only has 64 MB of DDR video, it gave performance equivalent to the Jetway 5200 128 MB. This shows that the quality of the components in this card is very good and the card is well worth the money. The XFX FX 5200 video card with 128 MB RAM performed worse than its 64 MB counterpart,

ATi vs nVidia on DirectX 9 Performance

Most of the tests that we ran, in this comparison, are DirectX 8 tests; and the nVidia cards beat ATi by a long shot in all of them. However, Valve Software has released benchmark figures for their first person shooter DirectX 9 game, *Half-Life 2*. These figures show ATi GPUs outperforming their nVidia counterparts by a significant margin.

A point to be noted here is that Valve has an OEM deal with ATi to provide their games with the latter's graphic card offerings. Valve, however, counters this by saying that they chose ATi as their GPUs are better than nVidia; as proved by the benchmarks. nVidia, on the other hand, claims that the Valve figures are invalid due to the fact that Valve used Release 45 drivers. They claim that the new detonator 50 drivers-currently in beta-will cover the performance gap, between nVidia and ATi, on *Half-Life 2* and other such shader-intensive games. The new drivers intelligently switch between DirectX 8 and DirectX 9, depending upon the instructions provided, without affecting image quality. To what extent this is true can only be determined after the drivers are released. Until then, what are the implications for prospective buyers? Well, nVidia cards are certainly better than ATi as far as the current crop of games go, but we recommend ATi to those who want to enjoy the games that will release in the near future.